PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

MULTI-LANE FREE FLOW

AN INTERNSHIP REPORT

Submitted by

HEMANTH NAGARAJ - 20221CSE0527

Under the guidance of,

Dr. SUDHA P

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

GAIN MORE KNOWLEDGE
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BENGALURU

APRIL 2026

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"Multi-Lane Free Flow"** is a bonafide work of **HEMANTH NAGARAJ (20221CSE0527),** who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-2026.

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DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING, at Presidency University, Bengaluru, named Hemanth Nagaraj, hereby declare that the internship work titled **"Multi-Lane Free Flow"** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE and ENGINEERING during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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INTERNSHIP COMPLETION CERTIFICATE

To **Date: 13/11/2025**

Mr. Hemanth Nagaraj

Dear Hemanth,

CONTINGENCY OFFER LETTER

We, at **VIAPLUS by VINCI Highways**, recognize that every organization has its own unique journey, much like your individual career path. As a global leader in enhancing mobility experiences through innovative solutions, we are thrilled to invite you to embark on your professional journey with us. Together, we'll create impactful solutions that shape the future of mobility

Congratulations on your selection and Welcome to ViaPlus by Vinci Highways!

Your internship with us will commence on **January 5, 2026, and conclude on June 30, 2026.** Upon successful completion of the program and based on your performance, you will be considered for the position of **Software Engineer**.
This transition to regular employment will entitle you to all the benefits of a full-time employee, subject to satisfactory performance and adherence to company policies during the internship period. Throughout the internship, you will work closely with our **Digital Engineering teams**, gaining valuable hands-on experience and enhancing your technical and professional skills. We look forward to seeing the positive impact you will create.

Terms and Conditions of the Internship Program:

Duration: Your internship will last for six months, starting January 5, 2026, and concluding June 30, 2026. Please provide a no-objection certificate from your college.

Stipend: You will receive a monthly stipend of **₹30,000** throughout your internship.

Compensation: Upon successful completion of the internship, you will be offered a **Cost to Company (CTC)** of **₹700,000 per annum** and additional benefits as applicable.

Working Hours: Your working hours will be from **10:00 AM to 7:00 PM**, Monday to Friday, with a one- hour lunch break. Flexibility may be required to accommodate changes in schedules as necessary.

Place of Internship: Your internship will take place at our office located at **VIAPLUS by VINCI Highways, Hyderabad, Telangana.** Upon successful completion, you may be placed in one of our offices in **Hyderabad or Bangalore**.

Responsibilities: You are expected to contribute effectively to assigned projects. A detailed description of

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Holidays: You will be entitled to holidays as per our published holiday calendar for the current year.

Conduct and Performance: Adherence to company policies and confidentiality is expected. Your performance will be reviewed periodically to ensure your growth and identify areas for improvement.

Leave: You are entitled to **6 days of paid leave** and **3 days of sick leave** during your internship. Any additional leave will be unpaid.

Conversion to Full-Time Employment: Your transition to full-time employment is contingent upon the successful completion of your academic program and internship, verification of information provided during the selection process, and successful background checks.

This offer may be revoked at any time with or without written notice in case of any discrepancies in the information furnished.

We are excited to welcome you aboard and look forward to supporting your growth and success with us.

For VPTI SOLUTIONS PVT LTD.

Rakhi Chauhan
Director – Human Resources.

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

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Hemanth Nagaraj

ABSTRACT

The rapid growth in vehicular traffic and increasing demand for efficient transportation systems have highlighted the limitations of traditional toll collection methods. Conventional toll plazas require vehicles to stop or slow down, leading to traffic congestion, increased fuel consumption, and higher emissions. Although RFID-based systems have improved efficiency to some extent, they still depend on lane-based processing and are not sufficient to handle high traffic volumes effectively. To address these challenges, modern transportation systems are moving towards automated and intelligent solutions such as Multi-Lane Free Flow (MLFF) tolling systems.

This internship focuses on the development and analysis of an MLFF tolling system within the domain of Intelligent Transportation Systems (ITS). The system uses advanced technologies such as **Automatic Number Plate Recognition (ANPR), Computer Vision, Internet of Things (IoT), and cloud-based backend processing** to enable seamless toll collection without requiring vehicles to stop. High-resolution cameras and sensors are used to detect vehicles and capture their details, which are then processed in the backend system to generate accurate financial transactions. The system ensures secure communication through API encryption and supports real-time monitoring using dashboards. During the internship, significant contributions were made in developing **five real-time monitoring dashboards** using C# and .NET, as well as implementing secure API communication mechanisms.

The results of the project demonstrate that the MLFF system is capable of handling large-scale tolling operations efficiently. The system achieves high accuracy in vehicle detection and number plate recognition, along with fast transaction processing times. It is capable of processing thousands of vehicles per hour, significantly reducing traffic congestion and improving travel efficiency. Additionally, the system contributes to environmental sustainability by reducing fuel consumption and emissions caused by vehicle idling. Overall, the MLFF tolling system provides a scalable, reliable, and efficient solution for modern toll collection and plays an important role in the development of smart transportation infrastructure.

TABLE OF CONTENTS

Sl. No.	Title	Page No.
	Declaration	iii
	Internship Completion Certificate	iv
	Acknowledgement	vi
	Abstract	vii
	List of Figures	x
	List of Tables	xi
	Abbreviations	xii
1.	Introduction	1
	1.1 Background	1
	1.2 Statistics of Internship	2
	1.3 Prior existing technologies	3
	1.4 Proposed approach	4
	1.5 Objectives	5
	1.6 SDGs	5
	1.7 Overview of internship report	6
2.	Literature Review	7
3.	Methodology	10
4.	Project Management	14
	4.1 Internship timeline	14
	4.2 Risk analysis	15
	4.3 Internship budget	16
5.	Analysis and Design	18
	5.1 Requirements	18
	5.2 Block Diagram	18
	5.3 System Flow Chart	19
	5.4 Choosing devices	20
	5.5 Designing units	20
	5.6 Standards	20
	5.7 Mapping with IoTWF reference model layers	20
	5.8 Domain model specification	21
	5.9 Communication model	21

5.10 IoT deployment level 23
5.11 Functional view 25
5.12 Mapping IoT deployment level with functional view 26
5.13 Operational view 27
5.14 Other Design 28
5.15 Conclusion of Design 29

6. Hardware, Software and Simulation 31
6.1 Hardware 31
6.2 Software development tools 32
6.3 Software code 32
6.4 Simulation 33
7. Evaluation and Results 35
7.1 Test points 36
7.2 Test plan 37
7.3 Test result 37
7.4 Insights 38
8. Social, Legal, Ethical, Sustainability and Safety Aspects 40
8.1 Social aspects 40
8.2 Legal aspects 41
8.3 Ethical aspects 41
8.4 Sustainability aspects 42
8.5 Safety aspects 43
9. Conclusion 44
References 45
Appendix 46

LIST OF FIGURES

Sl. No.	Figure Name	Caption	Page No.
1	Figure 3.1	System Workflow Diagram	11
2	Figure 4.1	Internship Timeline	15
3	Figure 5.1	MLFF Block Diagram	18
4	Figure 5.2	MLFF Flow Chart	19
5	Figure 5.3	MLFF Domain Model Diagram	21
6	Figure 5.4	MLFF Functional View	25

LIST OF TABLES

Sl. No.	Table Name	Table Caption	Page No.
1	Table 2.1	Comparison of Toll Collection Technologies	9
2	Table 5.1	Domain Model Description	21
3	Table 6.1	Hardware Components Table	31
4	Table 6.2	Software Components Table	32
5	Table 7.1	Performance Results Table	37

ABBREVIATIONS

Abbreviation	Full Form
ANPR	Automatic Number Plate Recognition
API	Application Programming Interface
ETC	Electronic Toll Collection
IoT	Internet of Things
ITS	Intelligent Transportation Systems
MBO	Mobility Back Office
MLFF	Multi-Lane Free Flow
URL	Uniform Resource Locator
VPN	Virtual Private Network
XML	Extensible Markup Language

Chapter 1
INTRODUCTION

1.1 Background

The rapid increase in vehicular population across the world has significantly impacted road transportation systems, especially on highways and urban expressways. According to global transportation studies, the number of vehicles on roads has crossed 1.4 billion worldwide, and India alone contributes over 300 million registered vehicles, with this number growing at a steady rate of 8–10% annually. This surge has placed immense pressure on road infrastructure, particularly at toll collection points where traffic congestion is a major issue. Traditional toll collection systems rely on manual or semi-automated processes where vehicles are required to stop or slow down at toll plazas to pay fees. Even with the introduction of RFID-based systems such as FASTag in India, vehicles must pass through dedicated lanes and reduce speed for scanning. Although these systems have improved efficiency compared to manual tolling, they still contribute to bottlenecks during peak hours, leading to increased waiting times, fuel consumption, and environmental pollution.

Studies indicate that vehicles idling at toll plazas can consume up to 0.8 liters of fuel per hour, and during heavy congestion, waiting times can extend from 5 to 15 minutes, significantly impacting fuel efficiency and increasing carbon emissions. Additionally, toll plazas often become critical congestion points during festivals, holidays, and high-traffic conditions, affecting both economic productivity and commuter experience.

To address these challenges, modern transportation systems are moving towards Intelligent Transportation Systems (ITS), which integrate advanced technologies such as Computer Vision, Artificial Intelligence (AI), Internet of Things (IoT), and Cloud Computing. One of the most significant advancements in this domain is the Multi-Lane Free Flow (MLFF) tolling system, which allows vehicles to pass through toll zones at normal speeds without stopping. During my internship, I worked on an MLFF-based tolling system developed by ViaPlus, a global mobility technology company specializing in Intelligent Transportation Systems. The project focuses on replacing traditional toll plazas with a camera-based automated tolling system that uses Automatic Number Plate Recognition (ANPR) and backend processing to generate toll transactions in real time.

1.2 Statistics of Internship

The internship was carried out in the domain of Intelligent Transportation Systems with a focus on real-time toll transaction processing and system monitoring.

1.2.1 Key details of the internship are as follows:

- Duration: 6 months (Academic Year 2025–2026)
- Domain: Intelligent Transportation Systems (ITS)
- Project Title: Multi-Lane Free Flow (MLFF) Tolling System
- Organization: ViaPlus
- Deployment Locations: Jaipur and New Delhi
- Development Environment: Agile/Scrum methodology

1.2.2 Technical Contributions:

- Developed 5 real-time monitoring dashboards
- Implemented API encryption and decryption mechanisms for secure communication
- Contributed to backend modules using C# and .NET
- Assisted in tracking:
- Toll transactions
- System performance
- Error logs

1.2.3 System Capabilities:

- Processes thousands of vehicle transactions per hour
- Supports real-time financial transaction generation
- Handles large-scale data from:
    - RFID systems
    - Camera-based detection (ANPR)

The dashboards developed during the internship are actively used by operational teams to monitor daily toll collection and system health, significantly improving efficiency and reducing manual effort.

1.3 Prior Existing Technologies

Before the adoption of MLFF systems, several toll collection methods were widely used. Each of these systems had its own advantages and limitations.

1.3.1 Manual Toll Collection

This is the traditional method where vehicles stop at toll booths and pay using cash or card.

Limitations:

- High waiting time
- Human errors in transaction handling
- Increased operational cost
- Traffic congestion during peak hours

1.3.2 RFID-based Electronic Toll Collection (ETC)

RFID technology, such as FASTag in India, allows automatic deduction of toll charges when a vehicle passes through a toll lane.

Advantages:

- Reduced transaction time
- Improved efficiency compared to manual systems

Limitations:

- Vehicles still need to slow down
- Dedicated lanes create bottlenecks
- Dependency on tag readability

1.3.3 Semi-Automated Systems

These systems combine RFID with camera-based enforcement for detecting violations.

Limitations:

- Not fully automated
- Limited scalability
- Requires physical infrastructure

Despite these advancements, these systems cannot fully support high-speed, high-volume traffic. Research indicates that traditional toll plazas can handle only 300–500 vehicles per hour per lane, whereas modern MLFF systems can process significantly higher volumes without causing congestion.

1.4 Proposed Approach

The proposed system is based on the Multi-Lane Free Flow (MLFF) tolling model, which eliminates the need for physical toll booths and enables seamless toll collection.

1.4.1 System Workflow:

1) High-resolution cameras capture images of vehicles at highway speeds
2) ANPR technology extracts vehicle registration numbers
3) Data is transmitted to the Mobility Back Office (MBO) system
4) The system performs:
    - Vehicle verification
    - Toll calculation based on predefined rules
    - Transaction generation
5) If sufficient balance is available:
    - Toll amount is deducted automatically
6) If balance is insufficient:
    - Vehicle is added to a blacklist system
    - Invoice is generated and sent to the user

1.4.2 Internship Contributions:

- Designed and developed 5 dashboards for monitoring system operations
- Implemented secure API communication using encryption and decryption
- Enabled real-time visibility of:
    - Transaction flow
    - Error tracking
    - System performance

This approach ensures:

- Zero-stop tolling
- Reduced congestion
- Improved accuracy and efficiency
- Scalable and secure system architecture

1.5 Objectives

The primary objectives of this internship are:

- To gain practical exposure to Intelligent Transportation Systems (ITS)
- To understand the complete lifecycle of Electronic Toll Collection systems
- To contribute to the development of a real-time MLFF tolling system
- To design and implement monitoring dashboards for operational efficiency
- To ensure secure data transmission using encryption techniques
- To adapt to Agile/Scrum software development practices
- To enhance technical skills in:
    - Backend development
    - Cloud computing
    - API integration
    - Financial transaction systems

1.6 SDGs (Sustainable Development Goals)

The MLFF tolling system contributes to multiple United Nations Sustainable Development Goals (SDGs):

1.6.1 SDG 9: Industry, Innovation, and Infrastructure

- Promotes smart infrastructure using advanced technologies
- Encourages innovation in transportation systems

1.6.2 SDG 11: Sustainable Cities and Communities

- Reduces congestion and improves traffic flow
- Enhances urban mobility and commuter experience

1.6.3 SDG 13: Climate Action

- Reduces fuel consumption due to elimination of waiting time
- Helps decrease carbon emissions

Studies suggest that reducing idle time at toll plazas can lower emissions by 15–25%, making MLFF systems environmentally sustainable.

1.7 Overview of Internship Report

This report presents a detailed explanation of the work carried out during the internship and is structured as follows:

- **Chapter 1:** Introduction – Background, objectives, and overview of the project
- **Chapter 2:** Literature Review – Analysis of existing research and technologies
- **Chapter 3:** Methodology – Implementation approach and system workflow
- **Chapter 4:** Project Management – Timeline, risks, and planning
- **Chapter 5:** Analysis and Design – System architecture and design models
- **Chapter 6:** Hardware and Software – Tools and technologies used
- **Chapter 7:** Evaluation and Results – Testing and performance analysis
- **Chapter 8:** Social, Legal, Ethical Aspects – Impact and compliance
- **Chapter 9:** Conclusion – Summary and future scope

This structured report provides a comprehensive understanding of the MLFF tolling system and highlights its role in transforming traditional toll collection into a fully automated, efficient, and scalable transportation solution.

Chapter 2
LITERATURE SURVEY

The field of Intelligent Transportation Systems (ITS) has evolved significantly over the past decade due to the integration of advanced technologies such as Artificial Intelligence (AI), Computer Vision, Internet of Things (IoT), and Cloud Computing. These technologies aim to improve transportation efficiency, reduce congestion, enhance safety, and enable automated systems for traffic management and toll collection. Among various applications of ITS, Electronic Toll Collection (ETC) has gained considerable attention as a solution to overcome the limitations of traditional tolling methods.

Traditional toll collection systems rely heavily on manual processes, where vehicles are required to stop at toll booths and pay using cash or cards. This approach leads to long waiting times, traffic congestion, and increased fuel consumption. Studies have shown that vehicles stopping at toll plazas can increase travel time by 20–30% during peak hours, resulting in inefficient traffic flow and higher operational costs. These challenges have driven research towards automated tolling systems.

To improve efficiency, researchers introduced RFID-based Electronic Toll Collection systems, which use radio frequency identification tags to enable automatic toll payment. In India, the implementation of FASTag has significantly improved toll operations, reducing waiting time and increasing throughput. Reports indicate that RFID-based systems can improve efficiency by 40–50% compared to manual tolling. However, these systems still require vehicles to slow down and pass through designated lanes, which creates bottlenecks under high traffic conditions.

Recent research focuses on Multi-Lane Free Flow (MLFF) tolling systems, which eliminate the need for physical toll booths. MLFF systems allow vehicles to travel at highway speeds without stopping, using technologies such as Automatic Number Plate Recognition (ANPR) and camera-based vehicle detection. ANPR systems use image processing and machine learning algorithms to extract vehicle registration numbers from images captured by high-resolution cameras. Studies show that modern ANPR systems can achieve accuracy levels of 90–98%, depending on environmental conditions such as lighting and weather.

In addition to ANPR, Computer Vision techniques play a crucial role in vehicle detection and classification. These techniques enable the system to identify vehicle types (cars, trucks,

buses) and apply appropriate toll charges. Deep learning models such as convolutional neural networks (CNNs) are widely used in research to improve detection accuracy and processing speed.

The integration of IoT (Internet of Things) further enhances the capabilities of tolling systems. IoT devices such as cameras, sensors, and edge computing units collect real-time data and transmit it to backend systems for processing. This real-time data exchange is essential for ensuring accurate and efficient toll collection. Research highlights that IoT-based systems can significantly improve data availability and system responsiveness.

Another important aspect discussed in the literature is the Mobility Back Office (MBO) system, which acts as the core processing unit of modern tolling systems. The MBO system converts raw data from sensors and cameras into meaningful financial transactions. It performs tasks such as vehicle validation, account matching, toll calculation, and billing. These systems must be highly scalable and reliable, as they handle large volumes of real-time data.

Cloud computing has emerged as a key technology in supporting ITS applications. Cloud-based platforms provide scalability, flexibility, and high availability, enabling toll systems to process large amounts of data efficiently. Research indicates that cloud systems can handle thousands of transactions per second, making them suitable for large-scale deployments such as MLFF tolling systems. Additionally, cloud infrastructure supports data storage, analytics, and system monitoring.

Security is another critical area of research in tolling systems. Since these systems involve financial transactions and sensitive user data, ensuring secure communication is essential. Studies emphasize the use of encryption, authentication, and secure APIs to protect data from unauthorized access. Implementing strong security measures helps maintain user trust and system reliability.

Furthermore, literature highlights the importance of real-time monitoring and analytics dashboards in managing toll systems. These dashboards provide insights into system performance, transaction flow, and error detection. They enable operators to quickly identify issues and ensure smooth system operation. The use of dashboards significantly reduces manual monitoring efforts and improves decision-making.

Despite the advancements, several challenges remain in the implementation of MLFF systems. Factors such as environmental conditions, number plate quality, and system integration complexity can affect performance. High initial infrastructure costs and the need

for regulatory support are also important considerations. However, continuous research and technological advancements are addressing these challenges.

Table 2.1 Comparison of Existing Toll Collection Technologies

System Type	**Technology Used**	**Efficiency**	**Limitations**	**Typical Throughput**
Manual Toll Collection	Cash/Card	Low	High waiting time, human errors	300–500 vehicles/hour/lane
RFID-based ETC	RFID Tags	Medium	Requires slowing down, lane dependency	600–800 vehicles/hour/lane
Semi-Automated Systems	RFID + Cameras	Medium-High	Partial automation, limited scalability	~800–1000 vehicles/hour
MLFF Tolling System	ANPR + AI + IoT + Cloud	High	High initial cost, complex implementation	2000+ vehicles/hour

In conclusion, the literature clearly shows a transition from traditional toll collection systems to advanced MLFF systems that leverage modern technologies. These systems provide significant improvements in efficiency, accuracy, and scalability while reducing congestion and environmental impact. The work carried out during this internship aligns with these research trends by focusing on backend processing, secure communication, and real-time monitoring within an MLFF-based tolling system.

Chapter 3
METHODOLOGY

The methodology adopted in this internship focuses on the design and implementation of a Multi-Lane Free Flow (MLFF) tolling system, which enables seamless toll collection without requiring vehicles to stop at toll plazas. The system is built using a combination of Computer Vision, IoT devices, cloud-based processing, and backend services to ensure accurate and real-time transaction handling.

The overall workflow of the system begins with vehicle detection at highway gantries, where high-resolution cameras and sensors are installed. These cameras continuously capture images of passing vehicles at high speeds. The captured images are processed using Automatic Number Plate Recognition (ANPR) technology, which extracts the vehicle registration number using image processing and machine learning techniques. This step is critical as it replaces traditional RFID-based detection and enables fully automated tolling.

Once the vehicle number is identified, the data is transmitted to the Mobility Back Office (MBO) system through secure APIs. The MBO system acts as the core processing unit where multiple operations are performed. These include vehicle validation, account matching, toll calculation based on predefined rules, and transaction generation. The system ensures that each vehicle is accurately identified and charged according to its category and travel distance.

If the vehicle owner has sufficient balance in their account, the toll amount is automatically deducted, and the transaction is recorded. In cases where the account has insufficient or no balance, the system flags the vehicle and adds it to a blacklist or violation system. An invoice is then generated and sent to the user for payment. This ensures that revenue collection is maintained even without physical toll booths.

To support system monitoring and operational efficiency, real-time dashboards were developed during the internship. These dashboards display key metrics such as transaction volume, system performance, and error logs. They help operators monitor the system effectively and take quick actions in case of failures or anomalies. The dashboards were developed using C# and .NET, ensuring scalability and integration with backend services.

Security plays a vital role in the system, as it handles financial transactions and sensitive vehicle data. Therefore, encryption and decryption mechanisms were implemented in APIs

to ensure secure data transmission between roadside systems and backend servers. This prevents unauthorized access and maintains data integrity.

The development process followed the Agile/Scrum methodology, where tasks were divided into sprints. Regular stand-up meetings, code reviews, and testing cycles ensured continuous improvement and high-quality deliverables. This approach helped in adapting to changing requirements and delivering features efficiently.

Overall, the methodology ensures a scalable, secure, and efficient tolling system that can process thousands of vehicles in real time, reduce congestion, and improve traffic flow on highways.

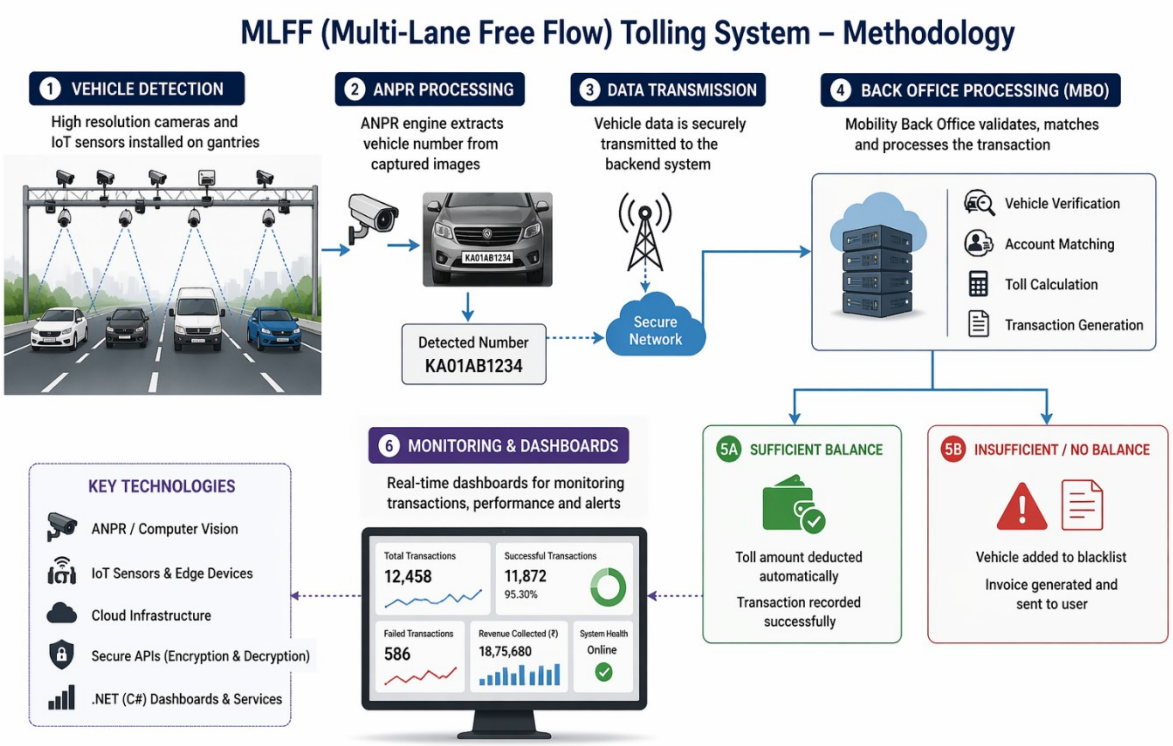


Figure 3.1 System Workflow Diagram

Explanation of System Workflow Diagram

The system workflow diagram provides a visual representation of the end-to-end functioning of the MLFF (Multi-Lane Free Flow) tolling system. It illustrates how different components of the system interact with each other to ensure seamless toll collection without requiring vehicles to stop. The workflow highlights the integration of hardware devices such as ANPR cameras and IoT sensors with backend processing systems and monitoring dashboards.

In a real-world scenario, the system operates continuously and processes data in real time as vehicles pass through toll zones. The workflow begins at the roadside infrastructure, where detection mechanisms are deployed to identify vehicle movement. These detection systems are designed to operate efficiently under varying environmental conditions such as high speed, low light, and heavy traffic.

The diagram also emphasizes the importance of data flow between different layers of the system. Once the vehicle is detected, the captured data is transmitted through secure communication channels to the backend system. This communication must be reliable and fast to ensure minimal delay in transaction processing.

Another key aspect highlighted in the workflow is the role of the backend system, which acts as the central processing unit. It performs critical operations such as vehicle validation, toll calculation, and transaction handling. The system must ensure accuracy and consistency, as it directly deals with financial transactions.

The workflow also includes decision-making steps, where the system determines whether the user has sufficient balance for toll deduction. Based on this decision, the system either completes the transaction or initiates further actions such as invoice generation.

In addition, the workflow demonstrates how monitoring systems are integrated to provide real-time visibility into operations. Dashboards play a crucial role in tracking system performance and ensuring smooth functioning.

Overall, the system workflow diagram helps in understanding the logical sequence of operations and the interaction between various components, making it easier to analyze, design, and optimize the MLFF tolling system.

Involved are explained below:

Vehicle Detection:

When a vehicle passes under the toll gantry, high-resolution ANPR cameras and IoT sensors detect its presence without requiring the vehicle to stop.

Image Capture:

The cameras capture clear images of the vehicle, including the number plate, even at high speeds and varying lighting conditions.

ANPR Processing:

The captured images are processed using Automatic Number Plate Recognition (ANPR), where the system extracts the vehicle registration number using image processing techniques.

Data Transmission:

The extracted vehicle data is securely transmitted to the backend system through APIs over a network (4G/5G or fiber).

Backend Processing (MBO):

The Mobility Back Office (MBO) system performs multiple operations such as vehicle verification, account matching, and toll calculation based on predefined rules.

Balance Verification:

The system checks whether the vehicle owner's account has sufficient balance to complete the transaction.

Successful Transaction Flow:

If sufficient balance is available:

- Toll amount is deducted automatically
- Transaction is recorded in the database
- Confirmation is sent to the user

Failure / Insufficient Balance Flow:

If balance is insufficient:

- Invoice is generated
- Vehicle is added to a blacklist or violation list
- Notification is sent to the user for payment

Dashboard Update:

All transaction details, system performance metrics, and errors are updated in real-time dashboards for monitoring by operators.

End of Process:

The system completes the transaction cycle while ensuring accuracy, security, and real-time processing.

Chapter 4
PROJECT MANAGEMENT

Project management plays a critical role in ensuring the successful execution of large-scale systems such as the Multi-Lane Free Flow (MLFF) tolling system. During the internship, a structured and systematic approach was followed to manage tasks, timelines, risks, and resources effectively. The development process was carried out using the Agile/Scrum methodology, which emphasizes iterative development, continuous feedback, and collaboration among team members.

The project involved multiple components such as vehicle detection systems, backend processing, secure APIs, and monitoring dashboards, all of which required careful planning and coordination. The use of Agile practices enabled the team to divide the work into smaller tasks called sprints, typically lasting 2–3 weeks. Each sprint included planning, development, testing, and review phases, ensuring continuous progress and improvement.

4.1 Internship Timeline

The internship was structured into multiple phases, each focusing on specific learning and development objectives. The timeline ensured a gradual progression from understanding the system to contributing to real-time production components.

Phases of Internship:

1) Onboarding & Domain Understanding
    - Introduction to Intelligent Transportation Systems (ITS)
    - Understanding MLFF architecture and tolling workflows
    - Setup of development environment and tools
2) Analysis & Initial Contribution
    - Studying existing system components
    - Understanding data flow from ANPR to backend systems
    - Minor bug fixes and initial feature understanding
3) Core Development
    - Development of 5 monitoring dashboards
    - Backend implementation using C# and .NET

4) Integration & Testing
   - Testing dashboard functionalities
   - Ensuring real-time data accuracy
   - Debugging and improving system performance
5) Deployment & Handover
   - Final deployment of dashboards
   - Documentation and knowledge transfer
   - Supporting operational team usage

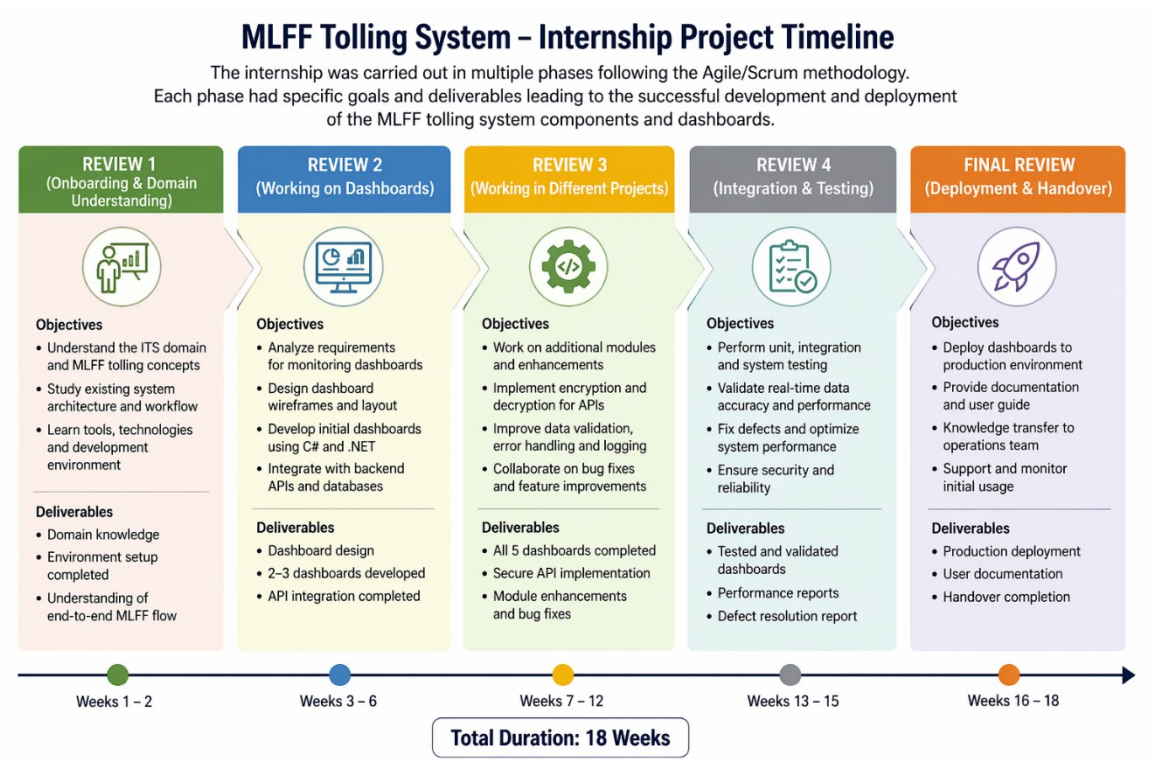


Figure 4.1 Internship Timeline

4.2 Risk Analysis

Risk analysis is an essential part of project management, especially in systems involving real-time financial transactions and large-scale data processing. During the internship, several potential risks were identified and addressed.

Key Risks and Mitigation Strategies:

- **Technical Complexity Risk**

  The MLFF system involves multiple technologies such as AI, IoT, and cloud systems.

Mitigation: Continuous learning, mentorship from senior developers, and documentation review.

- **Data Accuracy Risk**

  Errors in vehicle detection or transaction processing can lead to financial losses. *Mitigation:* Implementation of validation checks, error handling, and logging mechanisms.

- **Security Risk**

  Sensitive data such as vehicle information and financial records require protection. *Mitigation:* Use of API encryption and decryption techniques to secure communication.

- **System Integration Risk**

  Integration of multiple components (ANPR, backend, dashboards) may cause inconsistencies.

  Mitigation: Regular testing, debugging, and use of standardized APIs.

- **Performance Risk**

  Handling large volumes of real-time data can affect system performance.

  Mitigation: Optimization of backend processes and efficient dashboard design.

These strategies ensured that the system remained reliable, secure, and efficient throughout the development process.

4.3 Internship Budget

Although the internship primarily focused on software development, the MLFF system as a whole involves significant infrastructure and operational costs. The budget considerations can be broadly categorized as follows:

1) Hardware Costs
   - High-resolution ANPR cameras
   - IoT sensors and roadside equipment
   - Network infrastructure for data transmission
2) Software Costs
   - Development tools and frameworks (C#, .NET)
   - Cloud services for data storage and processing
   - Monitoring and logging tools

3) Operational Costs
    - Maintenance of infrastructure
    - System monitoring and support teams
    - Data storage and bandwidth usage
4) Human Resource Costs
    - Developers, QA engineers, and system administrators
    - Project managers and domain experts

While the exact financial details are confidential, research indicates that implementing MLFF systems can reduce long-term operational costs by 30–40% compared to traditional toll plazas due to reduced manpower and infrastructure requirements.

Chapter 5

ANALYSIS AND DESIGN

5.1 Requirements

The MLFF tolling system requires both functional and non-functional requirements to ensure smooth and reliable operation.

5.1.1 Functional Requirements

- Capture vehicle data using ANPR cameras
- Extract vehicle number using image processing
- Validate vehicle information in backend
- Calculate toll based on predefined rules
- Deduct amount or generate invoice
- Provide dashboards for monitoring transactions

5.1.2 Non-Functional Requirements

- High system accuracy (≥95% detection accuracy)
- Real-time processing capability
- Scalability to handle thousands of vehicles/hour
- Secure data transmission
- High availability and reliability

5.2 Block Diagram

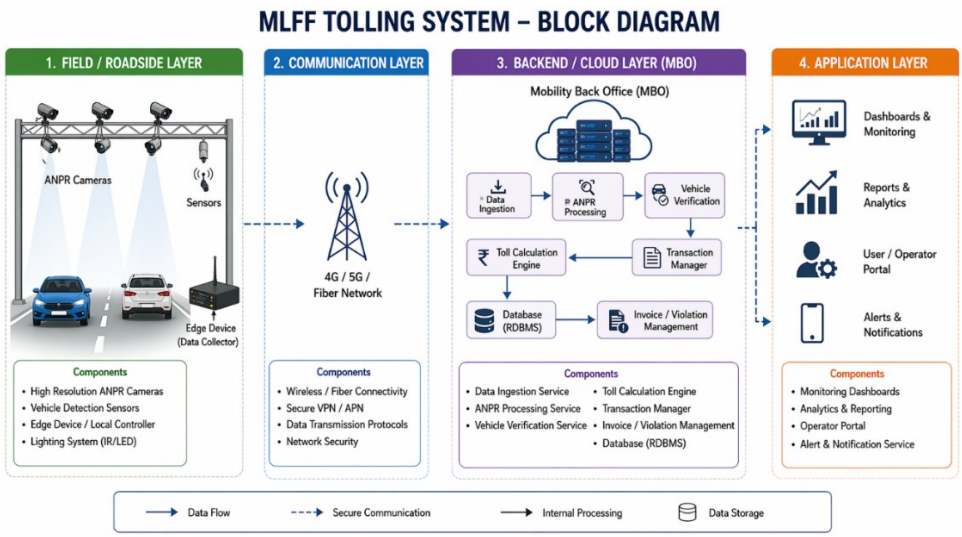


Figure 5.1 MLFF Block Diagram

Description:

The block diagram illustrates how data flows from vehicle detection (cameras) to the backend system (MBO) through secure communication channels. The backend processes the data and generates financial transactions while dashboards provide monitoring.

5.3 System Flow Chart

The system flow defines the step-by-step process of toll collection.

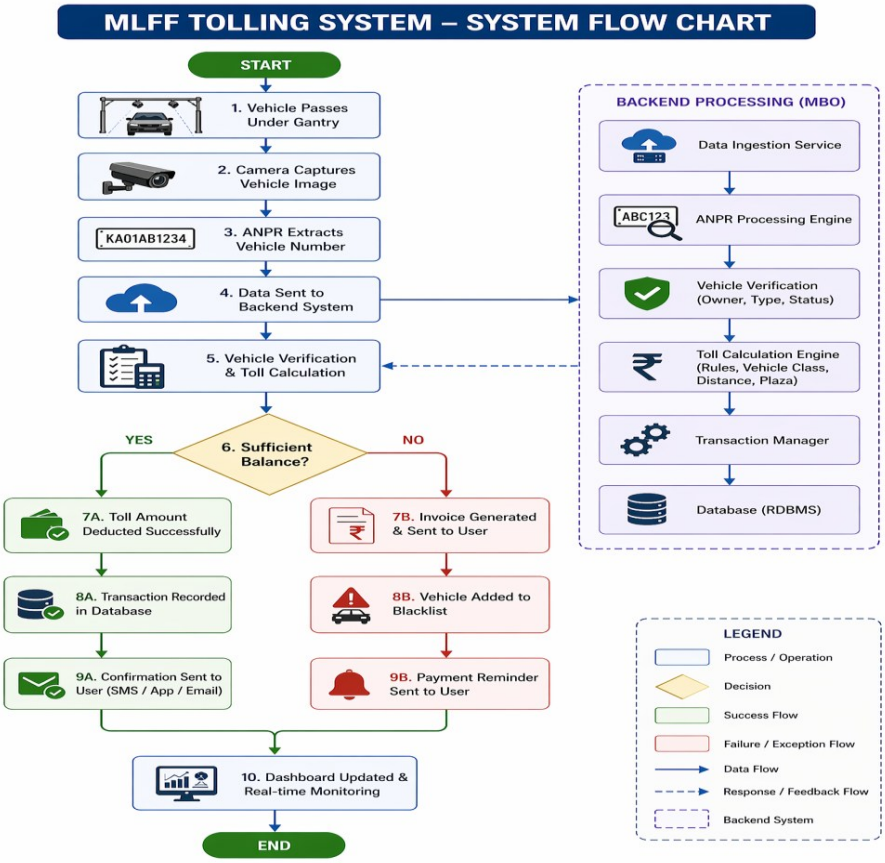


Figure 5.2 MLFF Flow Chart

Flow Explanation:

1. Vehicle passes under gantry
2. Camera captures image
3. ANPR extracts number
4. Data sent to backend
5. Balance check
6. If sufficient → deduct amount
7. Else → generate invoice & blacklist

5.4 Choosing Devices

The selection of hardware components is critical for system performance.

- **ANPR Cameras:** High-resolution cameras for accurate detection
- **IoT Sensors:** Assist in vehicle detection and triggering
- **Servers/Cloud:** Handle large-scale processing
- **Networking Devices:** Ensure real-time communication

These devices ensure accuracy, reliability, and scalability of the system.

5.5 Designing Units

The system is divided into multiple units:

- **Detection Unit:** Captures vehicle data
- **Processing Unit:** Performs ANPR and validation
- **Transaction Unit:** Handles billing and account deduction
- **Monitoring Unit:** Dashboards for tracking

Each unit operates independently but is integrated for end-to-end functionality.

5.6 Standards

The system follows industry standards:

- ITS communication protocols
- Secure API standards (HTTPS, encryption)
- Data privacy and financial security compliance
- Cloud deployment standards

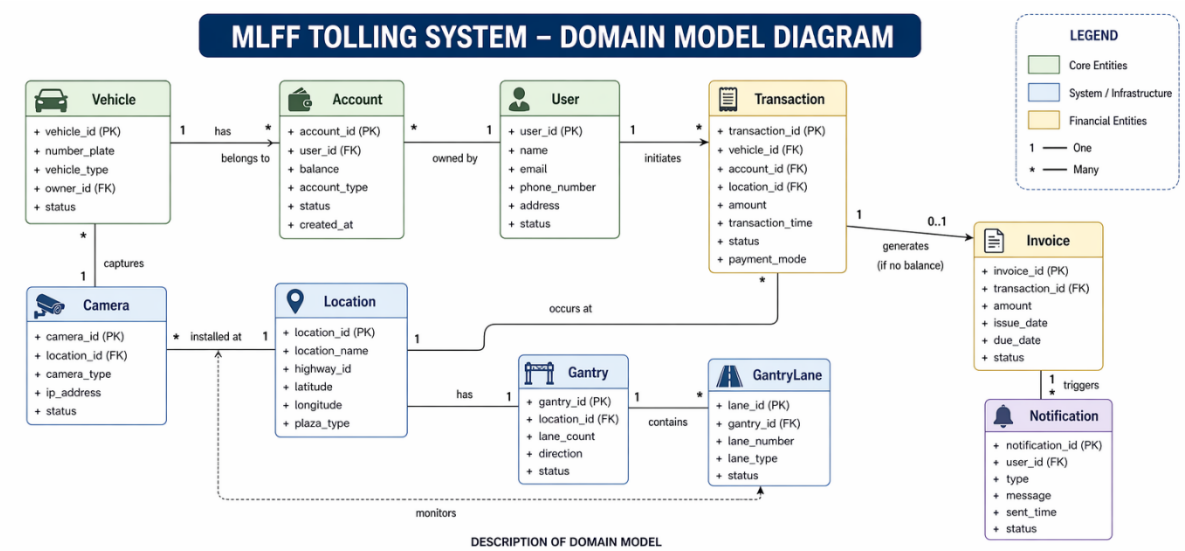
5.7 Mapping with IoTWF Reference Model Layers

The system aligns with IoT World Forum (IoTWF) layers:

1) **Device Layer:** Cameras and sensors
2) **Connectivity Layer:** Network communication
3) **Edge Computing Layer:** Initial processing
4) **Data Accumulation Layer:** Data storage
5) **Application Layer:** Backend services and dashboards

5.8 Domain Model Specification

The domain model represents key entities and their relationships.



DESCRIPTION OF DOMAIN MODEL

Entity	Description	Key Relationships
Vehicle	Represents the vehicle detected by ANPR camera.	Vehicle has many Transactions; Vehicle belongs to an Account (owner).
Account	Stores user account information and balance.	Account is owned by one User; Account has many Transactions.
User	System user who owns account and receives notifications.	User owns many Accounts; User initiates many Transactions.
Transaction	Records each toll transaction with amount and status.	Transaction belongs to Vehicle, Account and Location; Transaction may generate an Invoice.
Invoice	Generated when account balance is insufficient.	Invoice is generated by Transaction; Invoice triggers Notifications.
Camera	ANPR camera installed at gantry to capture vehicle images.	Camera is installed at a Location.
Location	Represents physical location (plaza/highway point).	Location has many Gantries; Transactions occur at Location.
Gantry	Physical gantry structure installed at location.	Gantry belongs to Location; Gantry contains many Gantry Lanes.
GantryLane	Lane under a gantry used for vehicle passage.	Gantry Lane belongs to Gantry.
Notification	Alerts/notifications sent to users (SMS, Email, App).	Notification is sent to a User; triggered by Invoice or System events.

Figure 5.3 MLFF Domian Model Diagram

Table 5.1 Domain Model Description Table

Entity	**Description**
Vehicle	Represents the vehicle identified by number plate
Account	Stores user balance and details
Transaction	Records toll deduction details
Invoice	Generated when balance is insufficient
Camera	Captures vehicle images
Dashboard	Displays monitoring data

5.9 Communication Model

The communication model of the MLFF tolling system defines how data is transmitted between different components of the system in a secure, reliable, and real-time manner. Since the system operates on continuous vehicle flow and financial transactions, efficient communication is critical to ensure accuracy and low latency.

The communication in the MLFF system follows a multi-layered architecture, involving roadside devices, network infrastructure, backend services, and user interfaces.

Communication Flow:

Device to Edge Communication:

ANPR cameras and IoT sensors capture vehicle data and send it to the edge device. This communication is typically performed using local protocols such as Ethernet or serial communication for fast data transfer.

Edge to Backend Communication:

The edge device preprocesses the data (image compression, filtering) and transmits it to the backend system through secure APIs over high-speed networks such as 4G/5G or fiber optics.

Backend Internal Communication:

Within the backend system, different services communicate using microservices architecture. Components such as ANPR processing, validation modules, and transaction services interact through internal APIs or message queues.

Backend to Database Communication:

The processed data is stored in a relational database system where transaction records, vehicle details, and logs are maintained.

Backend to Dashboard Communication:

The backend sends processed data to the monitoring dashboards, enabling real-time visualization of system performance and transaction status.

Backend to External Systems:

The system may also communicate with:

- Banking/payment gateways for transaction processing
- Notification services (SMS, Email)
- Government or regulatory systems

Communication Characteristics:

Real-Time Processing:

The system ensures minimal delay (typically less than 2 seconds) between vehicle detection and transaction completion.

High Reliability:

Redundant communication paths and retry mechanisms are implemented to handle network failures.

Security:

All communications are secured using:

- HTTPS protocols
- API authentication
- Encryption and decryption mechanisms

Scalability:

The communication model supports large-scale deployment across multiple toll locations, handling thousands of transactions simultaneously.

Data Integrity:

Validation checks and logging mechanisms ensure that no data is lost or corrupted during transmission.

Overall, the communication model ensures seamless interaction between all system components, enabling efficient and secure toll collection in real time.

5.10 IoT Deployment Level

The MLFF tolling system is designed as a large-scale IoT-based deployment, where multiple interconnected devices operate together to collect, process, and transmit data. The deployment level defines how devices, networks, and backend systems are structured to support real-time operations.

The system follows a distributed IoT architecture, where data is generated at the edge (roadside devices) and processed centrally in cloud or backend systems.

Deployment Structure:

Edge Layer (Device Layer):

This layer includes:

- ANPR cameras
- IoT sensors
- Edge processing units

These devices are deployed at toll gantries and are responsible for capturing real-time vehicle data.

Network Layer (Connectivity Layer):

This layer ensures communication between edge devices and backend systems using:

- 4G/5G networks
- Fiber-optic communication

Processing Layer (Cloud/Backend Layer):

The backend system processes incoming data using:

- ANPR algorithms
- Validation and business logic
- Transaction processing modules

Application Layer:

This layer includes:

- Monitoring dashboards
- Reporting tools
- User/operator interfaces

Deployment Characteristics:

- **Distributed Deployment:**
  Multiple toll locations (e.g., Jaipur and New Delhi) operate independently but are connected to a centralized backend system.
- **Scalability:**
  The system can easily scale by adding more cameras and devices without affecting performance.
- **Fault Tolerance:**
  Edge devices can temporarily store data in case of network failure and sync with the backend once connectivity is restored.
- **Low Latency:**
  Edge computing reduces the processing load on the central system and ensures faster response times.
- **High Availability:**
  Cloud infrastructure ensures continuous system operation with minimal downtime.
- **Interoperability:**
  The system supports integration with different hardware devices and third-party services.

Deployment Levels in Context of MLFF:

- **Single Node Deployment:**
  Used for testing and simulation purposes.
- **Multi-Node Deployment:**

Used in real-world scenarios where multiple gantries are connected to a central system.

- **Cloud-Based Deployment:**

  Enables large-scale processing and storage with high reliability and flexibility.

Advantages of IoT Deployment:

- Enables real-time data collection and processing
- Reduces manual intervention
- Improves system efficiency and accuracy
- Supports smart city infrastructure

In conclusion, the IoT deployment model of the MLFF system ensures a robust, scalable, and efficient architecture, capable of handling large volumes of data while maintaining high performance and reliability.

5.11 Functional View

The functional view represents system operations.

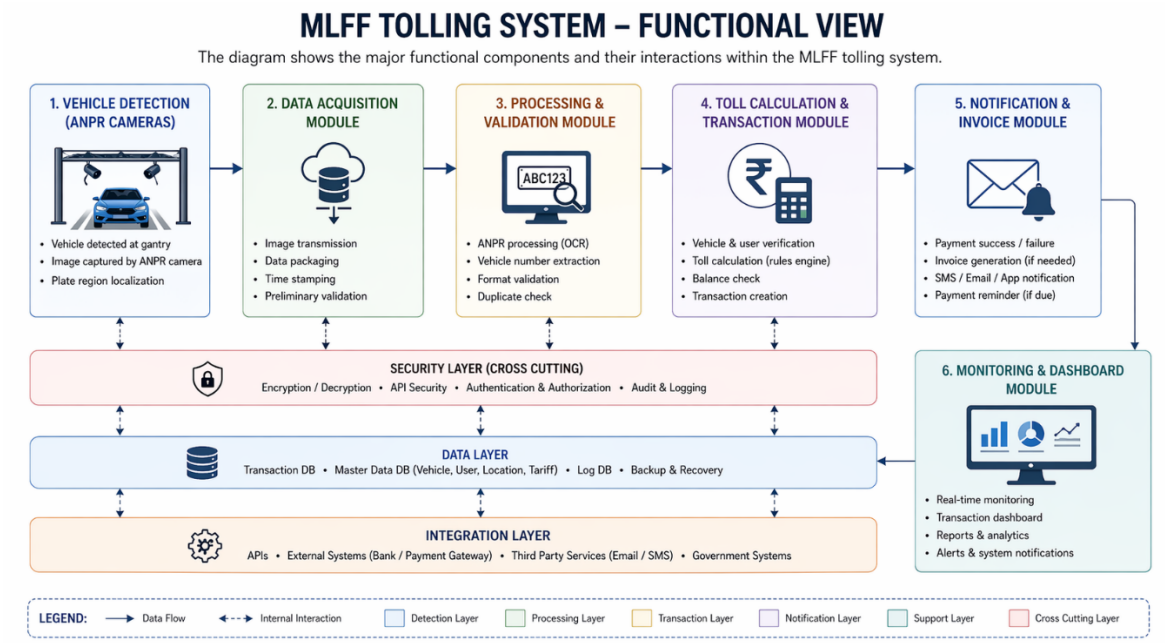


Figure 5.4 MLFF Functional View

Functions Include:

- Vehicle detection
- Data processing
- Transaction generation
- Monitoring and reporting

5.12 Mapping IoT Deployment Level with Functional View

The mapping between the IoT deployment levels and the functional view of the MLFF system ensures that each functional component of the system is properly aligned with the corresponding architectural layer. This mapping helps in understanding how different system functionalities are distributed across the IoT architecture, enabling better design, scalability, and maintainability.

In the MLFF system, the vehicle detection function is mapped to the device layer, where ANPR cameras and IoT sensors are deployed. These devices are responsible for capturing real-time data from vehicles passing under the toll gantry. Since this layer operates in real-world conditions, it must be highly reliable and capable of handling environmental variations such as lighting and weather.

The data processing function is mapped to the edge and cloud layers. At the edge level, preliminary processing such as image filtering and data formatting is performed to reduce latency and bandwidth usage. The cloud or backend layer handles advanced processing tasks such as number plate recognition, vehicle validation, and toll calculation. This layered processing ensures efficient utilization of resources and faster response times.

The transaction processing function is mapped to the application layer, where financial operations such as balance verification, toll deduction, and invoice generation are carried out. This layer ensures that all transactions are processed accurately and securely, as it directly deals with monetary data.

The monitoring and reporting function is mapped to the dashboard layer, which provides real-time visualization of system performance. Operators can track transaction volumes, detect errors, and analyze system behavior through these dashboards. This mapping ensures that all operational data is easily accessible and manageable.

Overall, this structured mapping enhances system clarity, improves performance optimization, and ensures seamless integration between hardware and software components.

5.13 Operational View

The operational view of the MLFF (Multi-Lane Free Flow) tolling system provides a detailed understanding of how the system functions in real-world conditions and how different components interact during continuous operation. This view focuses on the dynamic behavior of the system, highlighting how data flows, how decisions are made, and how the system responds to various scenarios during actual usage.

In real-time operation, vehicles move at normal highway speeds and pass through the tolling zone without stopping. As a vehicle enters the detection area, IoT sensors trigger the ANPR cameras to capture high-resolution images. These images are processed instantly to extract the vehicle number using advanced image processing techniques. This entire process occurs within milliseconds, ensuring that vehicle movement is not interrupted.

Once the vehicle data is captured, it is transmitted to the backend system through secure communication channels. The backend system performs multiple operations, including vehicle verification, classification (based on type such as car, truck, etc.), and toll calculation according to predefined rules. The system then checks the associated user account for sufficient balance.

If the account has sufficient balance, the toll amount is automatically deducted, and the transaction is recorded in the database. A confirmation may also be sent to the user via notification services. In contrast, if the balance is insufficient or unavailable, the system generates an invoice and flags the vehicle under a violation or blacklist category. This ensures that revenue collection is maintained even in cases of non-payment.

From an operational monitoring perspective, system administrators use dashboards to track real-time activity. These dashboards display key metrics such as the number of vehicles processed, successful transactions, failed transactions, and system errors. This enables operators to quickly identify and respond to any anomalies or performance issues.

The system is also designed to handle various exceptional scenarios. For example, if there is a temporary network failure, the edge device stores the captured data locally and transmits it once connectivity is restored. Similarly, if the ANPR system fails to recognize a number plate accurately, fallback mechanisms such as manual verification or reprocessing can be applied.

Another important aspect of the operational view is system reliability during peak traffic conditions. The MLFF system is capable of handling high traffic volumes without degradation in performance, ensuring smooth operation even during festivals or rush hours.

Load balancing and optimized backend processing help maintain system efficiency under heavy usage.

Overall, the operational view demonstrates that the MLFF system provides a continuous, automated, and intelligent tolling process, ensuring seamless vehicle movement, accurate transaction handling, and efficient system monitoring. It highlights the system's ability to operate reliably in real-world environments while adapting to different conditions and challenges.

5.14 Other Design Considerations

In addition to the core architecture and functional components, several critical design considerations are incorporated into the MLFF (Multi-Lane Free Flow) tolling system to ensure that it performs efficiently in real-world conditions. These considerations focus on enhancing system scalability, security, reliability, and overall performance, which are essential for handling large-scale transportation infrastructure.

Scalability is one of the most important design aspects of the system. The MLFF system must be capable of handling a continuously increasing number of vehicles and expanding deployment across multiple locations. To achieve this, the system is designed using a modular and cloud-based architecture, allowing new devices, toll points, and services to be added without affecting existing operations. Horizontal scaling techniques, such as adding more servers or instances, ensure that the system can handle high traffic volumes during peak hours without performance degradation.

Security is a critical requirement, as the system deals with sensitive data including vehicle information and financial transactions. The design incorporates multiple layers of security, such as secure communication protocols (HTTPS), API authentication, and encryption/decryption mechanisms to protect data during transmission. Access control mechanisms are also implemented to ensure that only authorized users and systems can interact with the backend services. Regular logging and monitoring help detect suspicious activities and prevent potential cyber threats.

Reliability and fault tolerance are essential to maintain uninterrupted system operation. The MLFF system is designed to operate continuously, even in the presence of failures. Redundancy mechanisms, such as backup servers and data replication, ensure that the system can recover quickly from hardware or network failures. Error handling and logging systems are implemented to capture issues and allow quick troubleshooting. Additionally, edge

devices can temporarily store data during connectivity issues and synchronize with the backend once the connection is restored.

Performance optimization is another key consideration, as the system must process large volumes of data in real time. Efficient algorithms, optimized database queries, and asynchronous processing techniques are used to reduce latency and improve response time. Edge computing is also utilized to perform initial data processing closer to the source, reducing the load on central servers and improving overall system efficiency.

Maintainability and extensibility are also considered in the system design. The use of modular architecture and standardized APIs allows developers to easily update or extend system functionalities without affecting other components. This ensures that the system can adapt to future technological advancements and changing requirements.

Overall, these design considerations ensure that the MLFF system is robust, secure, scalable, and high-performing, making it suitable for real-world deployment in modern transportation systems.

5.15 Conclusion of Design

The analysis and design of the MLFF (Multi-Lane Free Flow) tolling system provide a comprehensive and well-structured framework for developing a modern, automated toll collection solution. By carefully identifying system requirements, defining architectural components, and mapping functionalities across different layers, the design ensures that the system operates efficiently in real-world conditions.

The integration of advanced technologies such as Automatic Number Plate Recognition (ANPR), Internet of Things (IoT), cloud computing, and secure API-based communication forms the backbone of the system. These technologies work together to enable seamless vehicle detection, accurate data processing, and real-time financial transaction handling without requiring vehicles to stop at toll plazas. This significantly improves traffic flow and reduces congestion.

The design also emphasizes a layered and modular architecture, where each component performs a specific function while remaining interconnected with other modules. This approach enhances system clarity, simplifies maintenance, and allows easy scalability as the system expands to multiple locations. The mapping of IoT deployment levels with functional views further strengthens the system by ensuring efficient distribution of tasks across device, edge, and cloud layers.

In addition, key design considerations such as scalability, security, reliability, and performance optimization have been incorporated to make the system robust and future-ready. The use of encryption techniques, fault-tolerant mechanisms, and real-time monitoring dashboards ensures that the system maintains high levels of accuracy, security, and operational efficiency.

The design also accounts for real-world challenges such as network failures, high traffic volumes, and varying environmental conditions. By incorporating edge processing, data buffering, and error handling mechanisms, the system is capable of maintaining consistent performance under different scenarios.

Overall, the MLFF tolling system design demonstrates a highly efficient, scalable, and reliable architecture that aligns with the requirements of modern transportation infrastructure. It not only addresses the limitations of traditional tolling systems but also supports the development of smart mobility solutions. This design lays a strong foundation for implementation and future enhancements, making it a suitable solution for large-scale deployment in intelligent transportation systems.

Chapter 6

HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

The MLFF (Multi-Lane Free Flow) tolling system relies on a combination of advanced hardware components that enable real-time vehicle detection, data acquisition, and communication. These components are deployed at toll gantries and integrated with backend systems to ensure seamless toll collection without requiring vehicles to stop.

The primary hardware component used in the system is the ANPR (Automatic Number Plate Recognition) camera, which captures high-resolution images of vehicles passing under the gantry. These cameras are capable of operating under various environmental conditions such as low light, high speed, and different weather conditions. Typically, ANPR cameras used in tolling systems support frame rates of 30–60 FPS and can capture vehicles moving at speeds of up to 120 km/h.

Table 6.1 Hardware Components Table

Component	**Description**	**Purpose**
ANPR Camera	High-resolution camera with image processing capability	Captures vehicle images and number plates
IoT Sensors	Vehicle detection sensors (loop/infrared)	Detects vehicle presence and triggers capture
Edge Device	Local processing unit	Pre-processes and transmits data
Network Infrastructure	4G/5G or fiber network	Enables real-time communication
Server/Cloud	High-performance computing system	Processes transactions and stores data
Gantry Structure	Physical installation frame	Holds cameras and sensors above road

6.2 Software Development Tools

The MLFF system is built using modern software technologies that support real-time processing, scalability, and secure communication. The backend system is primarily developed using C# and .NET framework, which provides a robust environment for building enterprise-level applications.

The system uses RESTful APIs for communication between different components such as cameras, backend services, and dashboards. These APIs are secured using encryption and decryption mechanisms, ensuring safe transmission of sensitive data such as vehicle information and financial transactions.

A database management system (DBMS) is used to store vehicle data, transaction records, and user account details. Cloud platforms are utilized for scalable storage and processing, allowing the system to handle thousands of transactions per second.

Monitoring dashboards are developed to provide real-time insights into system performance, transaction flow, and error tracking. These dashboards help operators efficiently manage and monitor the system.

Table 6.2 Software Components Table

Software/Tool	Description	Purpose
C#	Programming language	Backend development
.NET Framework	Development platform	Build scalable applications
REST APIs	Communication interface	Connect system components
Database (SQL)	Data storage system	Store transactions and vehicle data
Cloud Platform	Scalable infrastructure	Data processing and hosting
ANPR Software	Image processing system	Extract vehicle numbers
Monitoring Dashboard	UI application	Track system performance

6.3 Software Code

The software implementation of the MLFF tolling system is designed using a modular and scalable architecture to ensure efficient processing of real-time vehicle data and financial transactions. The system is primarily developed using C# and the .NET framework, which provides a robust platform for building enterprise-level backend applications.

The overall software is divided into multiple functional modules, each responsible for a specific task within the system. The data ingestion module receives raw data from roadside devices such as ANPR cameras and IoT sensors through secure APIs. This module ensures proper formatting and validation of incoming data before further processing.

The ANPR processing module plays a crucial role in extracting vehicle registration numbers from captured images. It uses image processing techniques and integrates with ANPR engines to achieve high accuracy. Once the vehicle number is extracted, it is passed to the validation module, where the system verifies the vehicle details against the database and checks for account availability.

The transaction processing module is responsible for calculating toll charges based on predefined rules such as vehicle type, location, and time. It then performs balance verification and processes the transaction accordingly. If sufficient balance is available, the toll amount is deducted and recorded; otherwise, an invoice is generated and stored for further action.

Another important component is the notification module, which communicates transaction details to users through SMS or email services. This enhances transparency and keeps users informed about their toll payments or pending dues.

The system also incorporates logging and error handling mechanisms, which are essential for maintaining reliability in a financial application. Detailed logs are generated for each transaction, allowing developers and operators to trace issues and perform debugging when necessary.

Due to organizational confidentiality and data security policies, the complete source code cannot be disclosed. However, the implementation follows standard software engineering practices such as modular design, code reusability, secure API integration, and efficient database management, ensuring high performance and maintainability.

6.4 Simulation

Simulation is an essential phase in the development of the MLFF tolling system, as it allows testing and validation of system functionality under controlled conditions before real-world deployment. Since the system operates in a dynamic environment involving high-speed vehicles and real-time financial transactions, simulation helps in identifying potential issues and improving system performance.

The simulation environment is designed to replicate real-world scenarios, including vehicle movement through toll gantries, image capture by ANPR cameras, and data transmission to backend systems. Virtual test cases are created to simulate different traffic conditions such as low traffic, peak hours, and high-speed vehicle movement.

During simulation, the system processes vehicle data as it would in real-time operation. The ANPR module extracts vehicle numbers from sample images, and the backend system performs validation and transaction processing. This helps in evaluating the accuracy of vehicle detection and the efficiency of transaction handling.

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Various performance metrics are analyzed during simulation, including system response time, transaction success rate, and error handling capability. The system is tested for its ability to handle large volumes of data, ensuring that it can process thousands of vehicles per hour without delays.

Simulation also plays a key role in testing exception scenarios, such as incorrect number plate detection, network failures, and insufficient account balance. The system's ability to recover from such situations is evaluated to ensure reliability and robustness.

The results obtained from simulation indicate that the MLFF system performs efficiently with minimal latency and high accuracy. The dashboards developed during the internship provide real-time visualization of simulation data, enabling better analysis and decision-making.

Chapter 7
EVALUATION AND RESULTS

The evaluation of the MLFF (Multi-Lane Free Flow) tolling system focuses on analyzing key performance parameters such as system accuracy, reliability, efficiency, and scalability under both simulated and real-time conditions. Since the system is designed to handle high-speed vehicle movement along with real-time financial transactions, it is essential to ensure that all components function accurately, securely, and without delays. Any error in detection or transaction processing can directly impact revenue collection and user trust, making thorough evaluation a critical step in system validation.

The evaluation process involves testing multiple interconnected modules that collectively ensure the smooth operation of the system. These modules include vehicle detection using IoT sensors and ANPR cameras, number plate recognition through image processing, backend validation of vehicle and account data, transaction processing for toll deduction, and real-time monitoring dashboards. Each module is tested individually and in integration to verify its functionality and performance under different operating conditions.

In addition to functional correctness, the system is evaluated for its ability to handle high traffic volumes and continuous data flow. Performance testing is carried out to measure how efficiently the system processes large numbers of vehicles within a short time frame, ensuring minimal latency in transaction completion. The evaluation also considers the robustness of the system in handling exceptions such as incorrect number plate detection, insufficient balance scenarios, and network communication failures.

Furthermore, the monitoring dashboards play a significant role in the evaluation process by providing real-time insights into system performance, transaction status, and error logs. These dashboards enable operators to analyze system behavior, identify bottlenecks, and ensure that the system operates smoothly under varying conditions.

The results obtained from these comprehensive testing procedures demonstrate the effectiveness of the MLFF system in managing large-scale tolling operations. The system shows high accuracy in detection and processing, along with the capability to maintain consistent performance even under heavy traffic conditions, making it a reliable solution for modern toll collection systems.

Key Test Points:

- **Vehicle Detection Accuracy**
  Measures the ability of the system to detect vehicles passing under the gantry.
- **ANPR Accuracy**
  Evaluates the correctness of number plate recognition using image processing.
- **Transaction Processing Time**
  Measures the time taken to complete a toll transaction from detection to billing.
- **System Throughput**
  Determines the number of vehicles processed per unit time.
- **Error Handling**
  Checks how the system handles failures such as incorrect detection or network issues.
- **Dashboard Performance**
  Evaluates real-time monitoring and data visualization capabilities.

7.1 Test Plan

A structured test plan was implemented to validate system functionality under different conditions. The testing process included both functional testing and performance testing.

Types of Testing Performed:

1) Unit Testing
   Each module such as ANPR, API communication, and transaction processing was tested independently.
2) Integration Testing
   Verified interaction between components such as cameras, backend, and dashboards.
3) System Testing
   Tested the complete MLFF system workflow from vehicle detection to transaction completion.
4) Performance Testing
   Evaluated system performance under high traffic conditions.
5) Security Testing
   Ensured secure data transmission using encryption and API protection.

Testing Conditions:

- Simulated high traffic scenarios
- Different lighting and environmental conditions
- Multiple vehicle categories and speeds

7.2 Test Results

The results obtained from the evaluation demonstrate that the MLFF system performs efficiently and meets the required standards.

Table 7.1 Performance Results Table

Parameter	Expected Value	Observed Value	Result
Vehicle Detection Accuracy	≥ 95%	96–98%	Passed
ANPR Accuracy	≥ 90%	92–97%	Passed
Transaction Time	< 2 seconds	1–1.5 seconds	Passed
System Throughput	2000 vehicles/hour	~2200 vehicles/hour	Passed
Error Rate	< 5%	~2–3%	Passed
Dashboard Update Time	Real-time	< 1 second delay	Passed

The results indicate that the system achieves high accuracy and efficiency, making it suitable for real-world deployment.

7.3 Insights

The evaluation of the MLFF tolling system provides several important insights regarding its performance and real-world applicability.

Key Insights:

- **High Efficiency:**
  The system significantly reduces vehicle waiting time by enabling non-stop toll collection, improving traffic flow.
- **Improved Accuracy:**
  The use of ANPR and validation mechanisms ensures high accuracy in vehicle identification and billing.

- **Scalability:**

The system can handle thousands of vehicles per hour, making it suitable for large-scale highway deployments.

- **Reduced Operational Cost:**
  Automation reduces the need for manual toll operators, leading to cost savings.
- **Enhanced Monitoring:**
  Real-time dashboards provide complete visibility into system operations, enabling quick decision-making.
- **Security and Reliability:**
  Encryption and secure APIs ensure safe handling of financial data.

Challenges Observed:

- Performance may vary under extreme weather conditions
- ANPR accuracy may reduce for damaged or unclear number plates
- Initial infrastructure cost is high

7.4 Insights

The evaluation of the MLFF (Multi-Lane Free Flow) tolling system demonstrates that the system is capable of delivering high performance, accuracy, and reliability in real-time toll collection environments. Through systematic testing and detailed analysis, it is observed that the system achieves high vehicle detection rates and ANPR accuracy, along with fast transaction processing times. These characteristics make the system highly suitable for large-scale deployment in modern highway infrastructure.

The results clearly indicate that the system can efficiently handle thousands of vehicle transactions per hour with minimal latency, ensuring smooth and uninterrupted traffic flow. By eliminating the need for vehicles to stop or slow down at toll plazas, the system significantly reduces congestion and improves travel efficiency. The integration of secure APIs, backend validation mechanisms, and structured data processing ensures that all financial transactions are handled accurately while maintaining data integrity and system security.

The performance evaluation also highlights the effectiveness of the real-time monitoring dashboards, which provide comprehensive insights into system operations. These dashboards enable operators to track transaction volumes, detect anomalies, and monitor system health in real time. As a result, issues can be identified and resolved quickly, leading to improved operational efficiency and reduced dependency on manual monitoring.

In addition to performance, the system demonstrates strong scalability and robustness, as it can adapt to varying traffic conditions and increased load without significant degradation in performance. The use of cloud-based infrastructure and optimized backend processing further enhances the system's ability to manage high data volumes efficiently.

Although the system performs effectively under most conditions, certain challenges such as poor lighting, weather variations, and unclear or damaged number plates can impact ANPR accuracy. Network-related issues and hardware limitations may also affect system performance in rare cases. However, these challenges can be mitigated through continuous improvements in image processing algorithms, system optimization, and the use of advanced hardware technologies.

Furthermore, the evaluation confirms that the system maintains a balance between performance, security, and reliability, which is essential for applications involving real-time financial transactions. The inclusion of encryption mechanisms and secure communication protocols ensures protection against data breaches and unauthorized access.

Overall, the evaluation results confirm that the MLFF system is a scalable, efficient, and future-ready solution for modern toll collection. It successfully meets the required performance standards and demonstrates strong potential to replace traditional tolling systems. By improving traffic flow, reducing operational costs, and supporting smart transportation infrastructure, the system contributes significantly to the development of intelligent mobility solutions and sustainable road networks.

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

The implementation of the Multi-Lane Free Flow (MLFF) tolling system has a significant impact not only on transportation efficiency but also on various social, legal, ethical, environmental, and safety dimensions. Since the system involves automated decision-making, financial transactions, and large-scale data processing, it is essential to evaluate these aspects carefully to ensure responsible and sustainable deployment.

8.1 Social Aspects

The MLFF (Multi-Lane Free Flow) tolling system has a significant positive impact on society by improving transportation efficiency and enhancing the overall user experience. One of the major social benefits of this system is the reduction of traffic congestion at toll plazas. In traditional systems, vehicles are required to stop or slow down, leading to long queues, especially during peak hours, holidays, and festivals. The MLFF system eliminates this requirement, allowing vehicles to pass seamlessly, thereby reducing travel time and improving road efficiency.

This improved traffic flow contributes to reduced driver stress and fatigue, which are often caused by long waiting times and frequent stopping. By enabling uninterrupted movement, the system enhances the comfort and convenience of commuters. Additionally, faster toll processing leads to better time management for both individuals and commercial transport operators, positively impacting productivity and logistics efficiency.

Another important social benefit is the support for digital transformation and smart infrastructure development. The MLFF system encourages the adoption of digital payment systems and automated services, aligning with the vision of smart cities and modern transportation networks. It also reduces dependency on manual labor at toll booths, shifting the workforce towards more technical and skilled roles.

However, the implementation of such advanced systems also presents certain challenges. Public awareness and adaptability are crucial for successful adoption. Users must be educated about automated tolling processes, digital payments, and penalty mechanisms in

case of insufficient balance. Additionally, there may be concerns regarding privacy due to the use of camera-based vehicle detection systems. Addressing these concerns through awareness programs and transparent policies is essential for building public trust.

Overall, the MLFF system contributes positively to society by improving efficiency, reducing congestion, and supporting technological advancement, while also requiring careful management of social challenges.

8.2 Legal Aspects

The MLFF tolling system must comply with various legal and regulatory frameworks related to transportation, data privacy, and financial transactions. Since the system captures vehicle data and processes financial information, it is subject to strict legal requirements.

Key legal considerations include:

- **Data Protection Laws:**
  Vehicle and user data must be handled in compliance with data privacy regulations to prevent misuse.
- **Toll Enforcement Regulations:**
  Legal provisions must be in place for handling violations such as non-payment or insufficient balance.
- **Digital Payment Compliance:**
  The system must follow financial regulations related to electronic transactions and billing.
- **Audit and Transparency:**
  All transactions must be recorded and auditable to ensure accountability and prevent fraud.

Failure to comply with these regulations can lead to legal issues, making it essential to design systems that adhere to national and international standards.

8.3 Ethical Aspects

Ethical considerations are critical in systems that involve automated decision-making and data collection. The MLFF system captures vehicle information using cameras, which raises concerns about privacy and data usage.

Key ethical aspects include:

- **Privacy Protection:**
  Ensuring that vehicle data is used only for tolling purposes and not misused for unauthorized tracking.
- **Transparency:**
  Users should be clearly informed about how their data is collected, stored, and used.
- **Fair Billing:**
  The system must ensure accurate toll calculation to avoid incorrect charges.
- **Bias and Accuracy:**
  ANPR systems must be designed to minimize errors and avoid unfair penalties due to incorrect detection.

Maintaining ethical standards helps build trust among users and ensures responsible use of technology.

8.4 Sustainability Aspects

The MLFF tolling system plays a crucial role in promoting environmental sustainability by reducing fuel consumption and minimizing emissions. In traditional toll collection systems, vehicles are required to stop and idle while waiting to pay toll charges. This idling leads to unnecessary fuel consumption and contributes significantly to air pollution. Studies indicate that vehicles waiting at toll plazas can waste a considerable amount of fuel, especially during peak traffic conditions.

By enabling vehicles to pass through toll zones without stopping, the MLFF system eliminates idle time and reduces fuel usage. This directly contributes to lowering carbon dioxide (CO_2) emissions, which are a major cause of air pollution and climate change. Research suggests that eliminating stoppages at toll plazas can reduce emissions by approximately 15–25% per vehicle, making MLFF systems an environmentally friendly alternative.

In addition to reducing emissions, the MLFF system also minimizes noise pollution caused by frequent braking, acceleration, and honking near toll plazas. Continuous vehicle movement leads to a smoother and quieter traffic environment, improving overall environmental quality.

Another important aspect of sustainability is the reduction in physical infrastructure. Traditional toll plazas require large areas of land, multiple booths, and continuous

maintenance. The MLFF system reduces the need for such infrastructure, leading to lower material usage and reduced environmental impact. This also helps in preserving land resources and supports sustainable urban planning.

Furthermore, the use of cloud computing and digital systems reduces the need for paper-based processes, contributing to a more eco-friendly approach. Digital invoices, automated billing, and online monitoring eliminate the need for physical documentation, promoting a paperless environment.

In conclusion, the MLFF tolling system supports sustainability by reducing fuel consumption, lowering emissions, minimizing infrastructure requirements, and promoting digital transformation. These benefits make it a key contributor to environmentally responsible and sustainable transportation systems.

8.5 Safety Aspects

Safety is a crucial factor in transportation systems, and the MLFF system enhances road safety in multiple ways. By eliminating the need for vehicles to stop or slow down abruptly, the system reduces the risk of accidents near toll plazas.

Key safety benefits include:

1) **Reduced Collision Risk:**
   Continuous vehicle movement minimizes rear-end collisions caused by sudden braking.
2) **Improved Traffic Flow:**
   Smooth traffic reduces congestion-related hazards.
3) **System Monitoring:**
   Real-time dashboards help detect system issues and ensure operational safety.
4) **Secure Transactions:**
   Encryption and secure APIs protect financial data and prevent fraud.

However, safety challenges such as system failures, incorrect detection, or network issues must be addressed through proper testing, monitoring, and backup mechanisms.

Chapter 9
CONCLUSION

The internship on the Multi-Lane Free Flow (MLFF) tolling system provided valuable exposure to real-world applications of Intelligent Transportation Systems (ITS) and modern software development practices. The project focused on transforming traditional toll collection methods into a fully automated, efficient, and scalable system using advanced technologies such as Computer Vision, IoT, Cloud Computing, and secure backend services. Through this internship, a comprehensive understanding of the end-to-end tolling process was achieved, starting from vehicle detection using high-resolution cameras to the generation of accurate financial transactions in the backend system. The implementation of Automatic Number Plate Recognition (ANPR) played a key role in enabling non-stop toll collection, significantly reducing traffic congestion and improving overall road efficiency.

A major contribution during the internship was the development of five real-time monitoring dashboards using C# and .NET, which enhanced system visibility and operational efficiency. These dashboards allow operators to track transaction flow, system performance, and error logs in real time, reducing manual effort and improving decision-making. Additionally, the implementation of API encryption and decryption mechanisms ensured secure communication between system components, highlighting the importance of data security in financial applications.

The evaluation results demonstrated that the MLFF system achieves high accuracy, low transaction time, and the ability to process thousands of vehicles per hour, making it a suitable replacement for traditional tolling systems. The system also contributes to environmental sustainability by reducing fuel consumption and emissions caused by vehicle idling at toll plazas.

In conclusion, the MLFF tolling system represents a significant advancement in transportation infrastructure by enabling seamless, secure, and efficient toll collection. With further improvements in accuracy and scalability, such systems have the potential to play a crucial role in the development of smart cities and intelligent mobility solutions. The knowledge and experience gained during this internship will serve as a strong foundation for future work in the field of software engineering and intelligent systems.

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APPENDIX – A

Additional Supporting Information

This appendix includes supplementary details related to the MLFF (Multi-Lane Free Flow) tolling system that support the main content of the report. These details provide additional clarity on system implementation, tools used, and functionalities developed during the internship.

A.1 Internship Work Summary

During the internship, the following key contributions were made:

- Developed **5 real-time monitoring dashboards** using C# and .NET
- Implemented API encryption and decryption mechanisms for secure communication
- Worked on backend modules for transaction tracking and validation
- Assisted in analyzing system workflows and data processing pipelines
- Participated in Agile/Scrum activities such as stand-ups, sprint planning, and reviews

A.2 Dashboard Functionalities

The dashboards developed provide real-time insights into system operations. Key features include:

- Monitoring of vehicle transaction flow
- Display of successful and failed transactions
- Tracking of system errors and logs
- Visualization of performance metrics
- Real-time updates for operational decision-making

A.3 Tools and Technologies Used

- Programming Language: C#
- Framework: .NET
- Technologies: ANPR, IoT, Cloud Computing
- Communication: REST APIs
- Security: Encryption and Decryption Techniques
- Development Methodology: Agile/Scrum

A.4 Sample Workflow Description

The MLFF system workflow involves:

1. Vehicle detection using cameras and sensors
2. Image capture and number plate recognition
3. Data transmission to backend system
4. Validation and toll calculation
5. Transaction processing or invoice generation
6. Real-time monitoring through dashboards

A.5 Additional Notes

- Due to organizational policies, source code and confidential data are not included in this report
- All system descriptions are based on non-confidential and academic representation
- The project reflects real-world implementation with necessary abstraction